Retrieving Single Records

First() · FirstOrDefault() · Single() · SingleOrDefault()

1. The main difference

These LINQ methods return one entity instead of a collection. Choose based on whether a match must exist and whether duplicates are allowed.

Method	Expected result	No match	Multiple matches
First()	First matching row	Throws exception	Returns first match
FirstOrDefault()	First matching row, if any	Returns default (usually `null` for entities)	Returns first match
Single()	Exactly one row	Throws exception	Throws exception
SingleOrDefault()	Zero or one row	Returns default (usually `null` for entities)	Throws exception

Remember: FirstOrDefault() is useful when the record may not exist. Use Single() / SingleOrDefault() when uniqueness is required.

2. When to use each method

First() — when at least one row is guaranteed. Add `OrderBy()` if you need a predictable first row.

FirstOrDefault() — optional searches, such as looking up a student when no result is a normal possibility.

Single() — exactly one result must exist; missing or duplicate data is an error.

SingleOrDefault() — no result is acceptable, but duplicates are not, such as a unique email or course code.

3. SQL and logging

`ToQueryString()` previews the query shape. Runtime logging shows the SQL command actually executed by EF Core.

- **First()** commonly executes using `TOP(1)`.
- **Single()** commonly executes using `TOP(2)`, allowing EF Core to check whether a second result exists.

4. Short examples

FirstMethods.cs

```
var student = context.Students
    .Where(s => s.Status == "Active")
    .OrderBy(s => s.StudentId)
    .First();

var optionalStudent = context.Students
    .FirstOrDefault(s => s.Email == email);
```

SingleMethods.cs

```
var course = context.Courses
    .Single(c => c.Code == "EF-101");

var optionalCourse = context.Courses
    .SingleOrDefault(c => c.Code == "UNKNOWN-999");
```

5. Common mistakes

Mistake: using `First()` when no match may exist. It throws; use `FirstOrDefault()` and handle `null` when absence is expected.

Mistake: using `FirstOrDefault()` when duplicates signal invalid data. It hides the duplicate issue; use `SingleOrDefault()` instead.

Mental anchor: **uncertain existence** → **FirstOrDefault()**; **uniqueness matters** → **Single() / SingleOrDefault()**.

Next lesson: Find() vs FirstOrDefault() — Primary Key Retrieval.